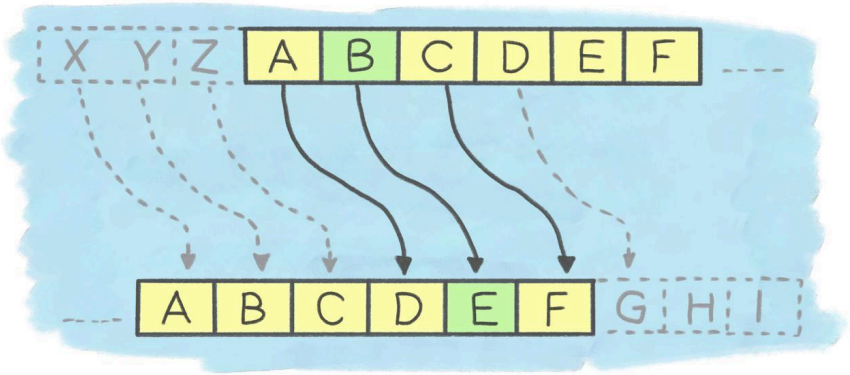
Caesar Cipher

Grade 3: What's in a Name?

Introduction

The need for secrecy when sending messages over the Internet is important for everyone. With this activity, students are introduced to the concept of basic cryptography/encryption, and to simple techniques for writing encrypted messages. Here, we will understand the Caesar cipher, use it to encrypt and decrypt a message. This activity is a very simple form of encryption used by Julius Caesar in the 1st century BCE.



Activity	Time	Description
Launch	5 min	Start by asking students if they have ever shared a secret message with a friend. Ask questions like: How did you make sure others could not understand it? or Did you ever use a code or secret language? The teacher can even write a sentence on board which is encrypted and the student has to guess the message. Tell students how long ago people also needed to send secret messages. Introduce the idea that even today, computers hide information using special codes to keep messages safe on the Internet. Explain that in this activity they will learn a simple way of hiding messages using a tool called the Caesar Cipher, and they will even make their own Caesar Cipher to send secret messages to their friends. Template: https://tinyurl.com/ceasersciphertemplate
Making & exploration of Caesar Cipher by Students	20 min	Give students the alphabet shifting table question. Making of Caesar Cipher Model and Student's Tryout the activity with each other following worksheet questions.

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		Student Worksheet: https://tinyurl.com/cearsercipherworksheet
Discussions & Conclusion	10 min	Discussing the worksheets. Students learned how to hide messages using a simple rule of shifting letters. They also discovered that while this method works, it is not very strong because there are only a few possible shifts. This shows why stronger encryption methods are needed to keep information safe today.

CT Components

Algorithmic Thinking
We follow a step-by-step process to encrypt and decrypt messages—choose a shift, match letters on the cipher wheel, and write the encoded letters.

Decomposition
We break the task of sending a secret message into smaller steps: choosing a message, deciding the shift, encoding each letter, and decoding it later.

Pattern Recognition
We can observe how letters consistently move forward in the alphabet by the same number of positions. By noticing this repeating pattern, they can quickly understand how the cipher works and even guess the shift used.

Generalisation
By trying different shift values (1, 2, 3, etc.), we can notice that the same rule works for any message or shift.

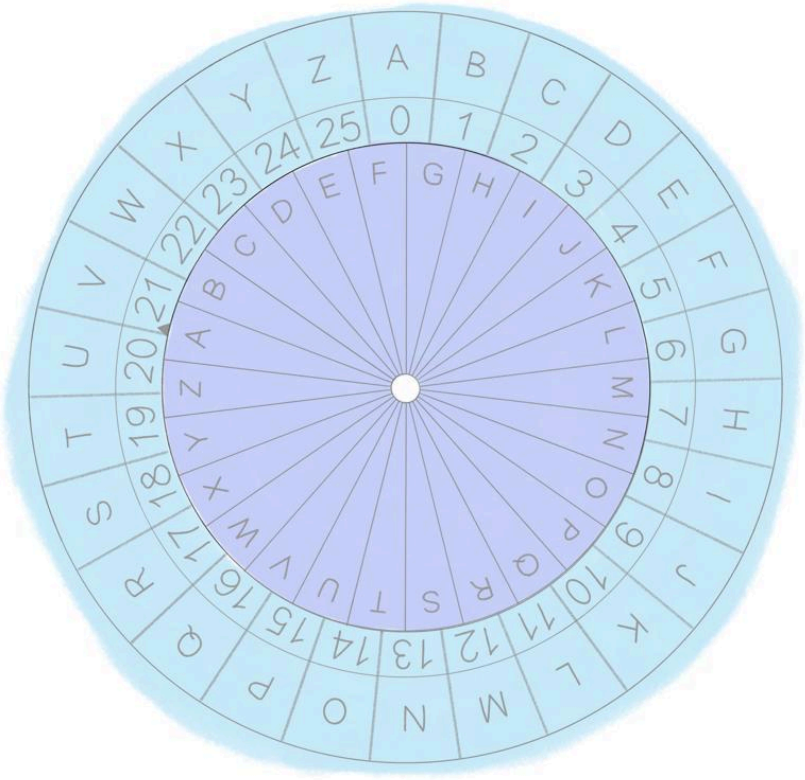
Evaluation
We can think about how secure the Caesar Cipher really is by exploring that there are only 25 possible shifts. This helps us to understand that some methods work but may not always be strong enough for real security.

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Activity: Caesar Cipher

People have used secret messages for centuries to protect important information. Even today, much of our communication online needs to be encrypted to keep our private data safe. For example, when your parents withdraw money from an ATM or make a UPI payment, they enter a PIN. This number needs to be hidden so that no one else can take their money, or when your parents buy something online and enter their bank card details, the information needs to be protected so that others cannot see or steal it.

A secret is information shared only with someone you completely trust through a message. Encryption is the process of changing a message so that its original meaning is hidden from anyone except the intended receiver. Though we use sophisticated methods to encrypt information today, we will understand encryption through a simple activity.



One simple way to do this is through ciphers. Cipher scrambles up the message using a **key**. To unscramble and access the message, you must have the key. This way, the cipher protects your message from those who do not have the key.

A well-known example of a cipher is the **Caesar Cipher**, named after the Roman leader Julius Caesar. He used this method to send important messages securely. In the Caesar Cipher, the letters of the alphabet are shifted by a certain number of places to be replaced by a new alphabet.

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For example, if the alphabet is shifted one place, A becomes B, B becomes C, and so on, while Z comes around to become A. This simple shifting method allows messages to be hidden from anyone who does not know the key, the shift number.

Q. Complete the table below by shifting the alphabet.

Original Placement	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Shift 1 place																										
Shift 2 place																										
Shift 20 place																										

Answer

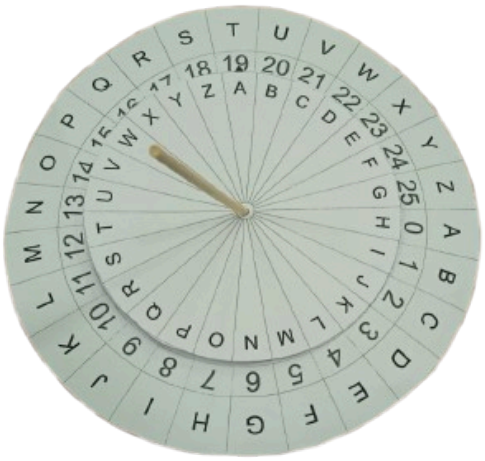
Original Placement	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Shift 1 place	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
Shift 2 place	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
Shift 20 place	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T

CT Competency :
Algorithmic Thinking
Pattern Recognition

Making of Caesar Cipher
Print out the template file on A4 Paper for each student.

(Guide Students to Make Caesar Cipher)

Cut out both the two circles, and carefully cut out the arrow on the smaller circle; it will help as an indicator. Put a toothpick or a thin stick through the centre of the two circles, with the smaller one on top of the larger one. Make sure that the small circle is free to rotate over the bigger circle.



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Q. If the message “**CAESAR CIPHER**” is encoded with a shift of 3 places, which of the following is the correct encoded message?

a) ECGUCT EKRJGT
b) FDHVDU FLSKHU
c) GEIWEV GMTLIV
d) DBFTBS DJQIFS

Explanation

b) FDHVDU FLSKHU
Each letter in the message is shifted in the alphabet by a fixed number of positions. Here, the shift is **3**: C → F, A → D, E → H, S → V, A → D, R → U
So CAESAR → FDHVDU, Similarly, CIPHER → FLSKHU

CT Competency :

Algorithmic Thinking
Decomposition
Generalisation
Pattern Recognition

Activity: Encrypting messages through Caesar Cipher

Make groups of two students. Once the students make the Caesar Cipher, tell them to write a message, encrypt it, and give it to their friends.

Choose a short message, like your name, your school name or the food item in your lunchbox that you would like to give to your friend, and decide on a number that will represent the shift. To use the cipher, first align the ‘A’ on the smaller circle with ‘A’ on the bigger circle. Now, turn the smaller circle such that the arrow lies on the chosen shift number.

Now, to encode (encrypt) your message, look for each letter of your message on the smaller circle, and write down the matching letter from the bigger circle. Do this one letter at a time until the entire message is encoded.

Once you have finished encoding your message, give your friend the encoded message and the shift letter (key). To decode (decrypt) the message, your friend should rotate the smaller circles so that the arrow matches the given key. Then your friend finds each letter of the encrypted message from the bigger circle and writes the corresponding letter on the smaller circle to reveal the original message.

Q. If the message “**THIS MESSAGE IS HIDDEN**” is encoded as “**ESTD XPDDLRP TD STOOPY**”, by how many positions has the alphabet been shifted?

a) 5

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b) 11
c) 17
d) 7

Explanation

b) 11

To find the shift, compare a letter from the original message to the corresponding letter in the encoded message. T → E, Moving 11 places from T in the alphabet comes around to E. Similarly, H → S, I → T, S → D. Each letter is shifted by 11 positions, confirming that the shift used here is 11.

CT Competency :

Algorithmic Thinking
Decomposition
Generalisation
Pattern Recognition

Q. How many possible meaningful ways are there to shift the alphabet in a Caesar cipher?

a) 26
b) 52
c) 25
d) 50

Explanation

c) 25

We can shift letters by a certain number of positions, determined by the number of letters in the alphabet. The English alphabet has 26 letters. But a Shift of 26 would bring A to itself, leaving the message unchanged. Therefore, only shifts from 1 to 25 produce encoded messages. So, there are 25 possible meaningful shifts.

CT Competency :

Generalisation
Evaluation

Think about how secure the Caesar Cipher really is by exploring that there are only a few possible shifts. This helps to understand that some methods work, but may not always be strong enough for real security. This may work for your secret messages to your friends, but not for your parents' bank transactions.

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